

Imanol G. Estepa

PhD Student at University of Barcelona

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EDUCATION

2023 – Present

PhD in Computer Vision

University of Barcelona

- Researcher (2023-2024) and PhD student (2024-Present) focused on Self-supervised Representation Learning and Image Generation, under the mentorship of **Dr. Bhalaji Nagarajan** and **Prof. Petia Radeva**, a **top 2% AI researcher** globally. See research profile [↗](#) and global ranking [↗](#).
- Recipient of the **prestigious FPU Fellowship** funded by the Spanish Ministry of Universities, given to most promising students across the country. 8th candidate in the country.

2021 – 2023

Master's Degree in Artificial Intelligence

Universitat Politècnica de Catalunya

- **Average grade:** 9.37/10
- Scores in courses relevant to my research: 9.9/10 in Computer Vision, 10/10 with honors in Unsupervised Learning.
- Master's thesis "*Analysis and improvement proposal on self-supervised deep learning*" graded 10/10 with honors (mention given in exceptional cases). The manuscript is publicly available [↗](#).

2017 – 2021

Bachelor's Degree in Computer Science

University of Deusto

- **Average grade:** 8.54/10
- Industry linked Bachelor's thesis that led to 3 different patents.
- Student representative for all bachelor years.

PUBLICATIONS

2025 **Estepa, I.G.**, Rodríguez-de-Vera, J.M., Sarasúa, I., Nagarajan, B., & Radeva, P. "Conjuring Positive Pairs for Efficient Unification of Representation Learning and Image Synthesis", *arXiv preprint arXiv:2503.15060*, 2025. Available on [\[Arxiv\]](#) [↗](#). UNDER REVIEW.

2025 Rodríguez-de-Vera, J.M., **Estepa, I.G.**, Nagarajan, B., & Radeva, P. "CLIP-DoRA: Weight-Decomposed Low-Rank Adaptation for Efficient Vision-Language Models", in the *International Neural Network Society Workshop on Deep Learning Innovations and Applications* located at the *International Joint Conference on Neural Networks*, 2025.

2024 Rodríguez-de-Vera, J.M., **Estepa, I.G.**, Sarasúa, I., Nagarajan, B., & Radeva, P. "Precision at Scale: Domain-Specific Datasets On-Demand", *arXiv preprint arXiv:2407.03463*, 2024. Available on [\[Arxiv\]](#) [↗](#). UNDER REVIEW.

Rodríguez-de-Vera, J.M., **Estepa, I.G.**, Bolaños, M., I., Nagarajan, B., & Radeva, P. "LOFI: LOnG-tailed Fine-Grained Network for Food Recognition", in 1st MetaFood Workshop in the *IEEE/CVF Conference on Computer Vision and Pattern Recognition 2024 (CVPR'24)*, Seattle, USA, June 17th, 2024

2023 **Estepa, I.G.**, Sarasúa, I., Nagarajan, B., & Radeva, P. "All4One: Symbiotic Neighbour Contrastive Learning via Self-Attention and Redundancy Reduction", in the *International Conference on Computer Vision 2023 (ICCV'23)*, July 2023.

Estepa, I.G., Rodríguez-de-Vera, J.M., Nagarajan, B., & Radeva, P. "Good Fences Make Good Neighbours", in *4th Visual Inductive Priors for Data-Efficient Deep Learning Workshop in International Conference on Computer Vision 2023 (ICCV'23)*, July 2023.

Rodríguez-de-Vera, J.M., Villacorta, P., **Estepa, I.G.**, Bolaños, M., Sarasúa, I., Nagarajan, B., & Radeva, P. "Dining on Details: LLM-Guided Expert Networks for Fine-Grained Food Recognition", in *8th International Workshop on Multimedia Assisted Dietary Management in the 31st ACM International Conference in Multimedia (ACMM'23)*, Ottawa, Canada, October 29th, 2023

CONFERENCES

June 2024 **IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR'24)** - Seattle, USA
Oral and poster presentation of "LOFI: LOng-tailed Fine-Grained Network for Food Recognition" in the 1st **MetaFood** Workshop.

October 2023 **ACM Multimedia 2023** - Ottawa, Canada
Oral and poster presentation of "Dining on Details: LLM-Guided Expert Networks for Fine-Grained Food Recognition" in the 8th International Workshop on **Multimedia Assisted Dietary Management**. Awarded a Volunteering Grant covering all expenses for this conference.

October 2023 **International Conference on Computer Vision 2023 (ICCV'23)** - Paris, France

- Poster "All4One: Symbiotic Neighbour Contrastive Learning via Self-Attention and Redundancy Reduction" in the **Main Conference**.
- Poster "Good Fences Make Good Neighbours" in the **4th VIPriors** workshop.
- Poster presentation of the extended abstract "BlendShift: Adaptive Neighbour Correction and Replacement for Efficient Neighbour Contrastive Learning" in LatinX Workshop.

2023 - Present **Regional Conferences and Symposiums** - Barcelona, Spain

- Regular participant in regional AI and Computer Vision symposiums (2022–2024), contributing presentations and posters on ongoing research.

AWARDS & DISTINCTIONS

2023-Present **FPU Fellowship**
Selected as the **top-ranked candidate** of the country for the FPU fellowship granted by the Spanish Ministry of Education and Science. This prestigious fellowship recognizes outstanding doctoral research potential and supports researchers pursuing their PhD studies in Spain. The fellowship includes a monthly stipend, funding for research stays, and a tuition waiver.

2023 **ICCV 2023 Volunteering Grant**
Awarded a Volunteering Grant to attend the ICCV 2023 Paris, France. This grant covered all expenses related to conference attendance.

2023 **European Project fellow**
Competitive fellowship awarded for European Project participation under the supervision of Petia Radeva.

WORK EXPERIENCE

March 2022 – March 2025 **Deep Learning Researcher and Research Intern**
AIBA research group, Dept of Mathematics and Computer Science, Universitat de Barcelona, Spain

- Part European and Regional projects: Measurer and Nuclis Deepsense.
- Produced several scientific publications.
- Consistently worked under the supervision of Dr. Petia Ivanova Radeva.

October 2019 – May 2021 **Student Researcher**
DeustoTech Technology Center , Basque Country, Spain

- Development of a Deep learning model to convey efficient predictive maintenance of industry blast furnaces. Patent registered.
- Implementation of an industrial dashboard to control a completely automatized production plant. Patent registered.
- Development of a complete AI suite for Taxi reservation prediction and demand management (Oporto, Portugal). Patent registered.

OUTREACH AND SERVICE

July 2025 **Summer School: Demystifying Artificial Intelligence** - Barcelona, Spain
Organizer and Lecturer. Developed and conducted a comprehensive lecture on GenAI ("*Generative AI: The Quest to Imitate the "Reality"*") and hands-on lab session for international engineering students at the Bachelor's, Master's, and PhD levels from partner institutions abroad.

July 2025 **IL3 Public Lectures - Barcelona, Spain**

Lecturer. Developed and conducted a comprehensive Spanish lecture novel Generative AI techniques and their impact in our daily life.

June 2025 **CVPR Workshop Challenge - Dishcovery: VLM MetaFood Challenge - Nashville, USA**

Organizer. Designed the evaluation task and created the training and test datasets. 274 submissions and more than 800 downloads of the training dataset during the lapse of the contest.

January 2025 **Winter School: Demystifying Artificial Intelligence - Barcelona, Spain**

Organizer and Lecturer. Developed and conducted a comprehensive lecture on GenAI ("*Generative AI: The Quest to Imitate the "Reality"*") and hands-on lab session for international engineering students at the Bachelor's, Master's, and PhD levels from partner institutions abroad.

July 2024 **IL3 Public Lectures - Barcelona, Spain**

Lecturer. Developed and conducted a comprehensive Spanish lecture on societal challenges of the Generative AI for non-experts.

SKILLS

- Languages** Spanish: native speaker.
Basque: native speaker, CEFR level C1/HABE - Advanced (Certificate issued by Helduen Alfabetatze eta Berreuskalduntzerako Erakundea).
English: CEFR level C1 - Advanced (Certificate issued by Cambridge Assessment English).
- Soft Skills** Public Speaking • Verbal Communication • Clarity of Thought • Adaptability • Collaboration